Hotel Demand Forecasting Workflow

Python, pandas, Prophet, matplotlib | Hotel Booking Demand dataset

1

Load the data

Read the hotel bookings CSV into pandas and inspect it with `head()`, `columns` and `shape`. The dataset has 119,390 rows and 32 columns.

```
df = pd.read_csv('hotel_bookings demand.csv')
```

2

Build the arrival date

Combine arrival year, month and day into one datetime column, `arrival_date`.

```
df['arrival_date'] = pd.to_datetime(year + '-' + month + '-' + day)
```

3

Aggregate to daily metrics

Group by arrival date to get daily demand (booking count), average ADR and total cancellations.

```
df.groupby('arrival_date').size()
```

4

Prepare and fit Prophet

Rename the daily demand to `ds` and `y`. Fit Prophet with yearly and weekly seasonality on and daily seasonality off.

```
Prophet(yearly_seasonality=True, weekly_seasonality=True)
```

5

Forecast 90 days

Extend the timeline by 90 days (to 29 Nov 2017), predict, and plot the forecast and its trend and seasonality components.

```
model.make_future_dataframe(periods=90)
```

6

Visualise performance

Plot daily demand, ADR, cancellations and revenue, plus a combined demand, ADR and cancellations chart.

```
plt.plot(daily_adr.index, daily_adr.values)
```